

Automata

languages lexers, parsers, interpreters



Introduction continued

Basic concepts (more formal)

- Language
 - Alphabet: nonempty set Σ of symbols
 - Language: subset of Σ^*
- Grammar
 - A means to study language mathematically
 - Definition of grammar $G = (V, T, S, P)$,
- Automata
 - an abstract model of a digital computer

Grammar

- $G = (V, T, S, P)$,
 - V is a finite set of objects called variables,
 - T is a finite set of objects called terminal symbols,
 - $S \in V$ is a special symbol called the start variable,
 - P is a finite set of productions.

The set of all such terminal strings is the language defined or generated by the grammar

Grammar sample

- $G = (\{A, S\}, \{a, b\}, S, P)$

With P given by:

- $S \rightarrow aAb \mid \lambda$
- $A \rightarrow aAb \mid \lambda$
- Some productions
 - $S \Rightarrow \lambda$
 - $S \Rightarrow aAb \Rightarrow ab$
 - $S \Rightarrow aAb \Rightarrow aaAbb \Rightarrow aaaAbbb \Rightarrow aaabbbb$
- $L(G) = \{a^n b^n : n \geq 0\}$

Automata

- An automation is an abstract model of a digital computer
- Automation can be described by
 - Grammar
 - By state transition diagram
 - Sample accepter accepts all legal C identifiers

